

Immunotherapy Immunophenotyping

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This document contains the analysis of immunophenotyping of tumor tissue from mice treated with Immunocheckpoint blockade and chemoimmunotherapy.

First, let's plot the data of tumor infiltrating lymphocytes

```
#Accessing the data folder
setwd("C:/Users/luisr/Documentos/Documents/Documentos - Copia/Mestrado/imunofenotipagem")

#Importing data
tilData <- read.csv("tils.csv")

head(tilData)
```

	group	CD3CD8	CD3CD4	CD3	exhaust
1	ICB+QI	22.90	23.20	34.6	28.6
2	ICB+QI	9.66	8.81	31.6	26.3
3	ICB+QI	9.13	9.35	27.4	22.0
4	ICB	11.60	14.90	26.2	47.6
5	ICB	6.95	7.52	19.6	32.4
6	ICB	1.82	2.02	18.5	22.2

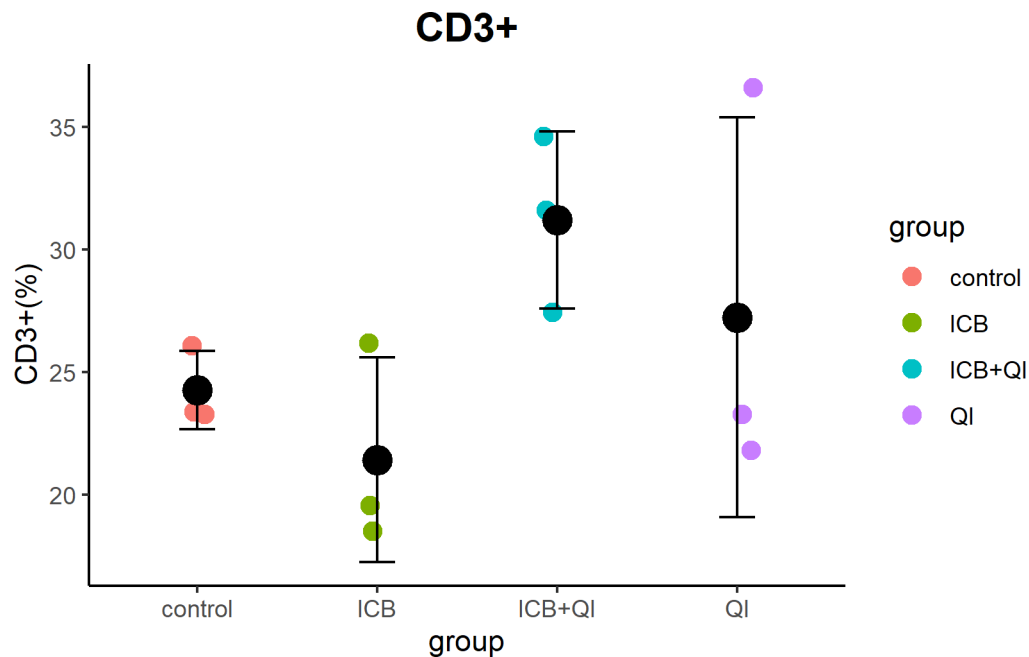
```
library(tidyverse)
library(magick)

tilPlot <- ggplot(tilData,
                  aes(
                    x = group,
                    y = CD3,
                    color = group
```

```

    )) +
  geom_jitter(
    width = 0.1,
    size = 3
  ) +
  stat_summary(
    fun = mean,
    geom = "point",
    size = 5,
    color = "black"
  ) +
  stat_summary(
    fun.data = mean_sdl,
    fun.args = list(mult = 1),
    geom = "errorbar",
    width = 0.2,
    color = "black"
  ) +
  theme_classic() +
  labs(
    title = "CD3+",
    y = "CD3+(%)"
  ) +
  theme(plot.title = element_text(
    hjust = 0.5,
    face = "bold",
    size = 15
  ))
)
tilPlot

```



```
ggsave(
  filename = "CD3.pdf",
  plot = tilPlot,
  width = 8,
  height = 6
)
```

Now, let's plot the TCD8+ lymphocytes

```
#Accessing the data folder
setwd("C:/Users/luisr/Documentos/Documents/Documentos - Copia/Mestrado/imunofenotipagem")

#Importing data
tilData <- read.csv("tils.csv")

head(tilData)
```

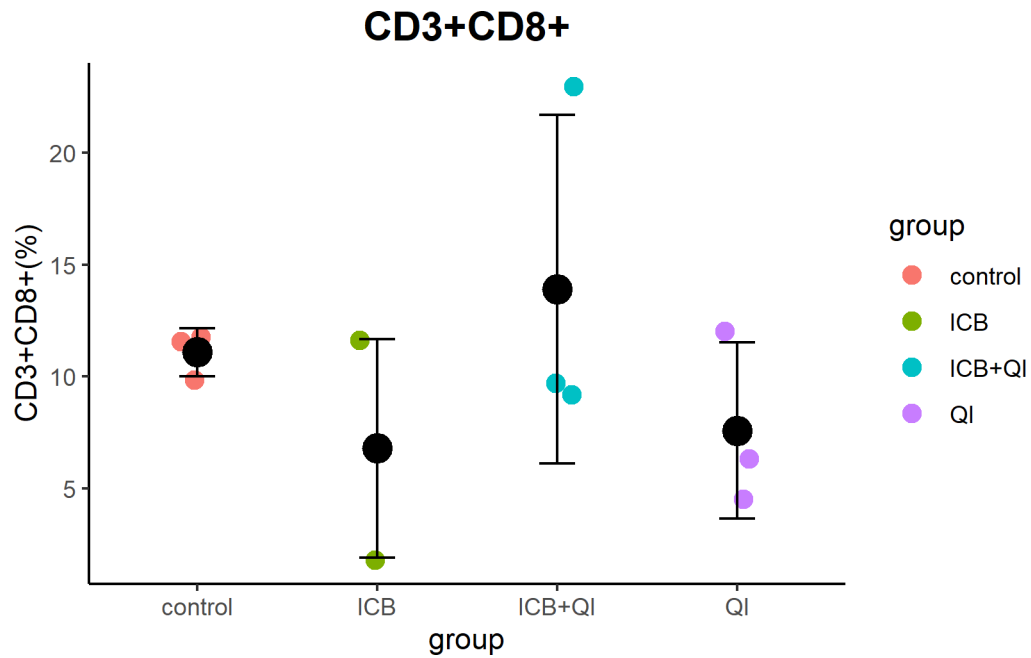
	group	CD3CD8	CD3CD4	CD3	exhaust
1	ICB+QI	22.90	23.20	34.6	28.6
2	ICB+QI	9.66	8.81	31.6	26.3
3	ICB+QI	9.13	9.35	27.4	22.0

4	ICB	11.60	14.90	26.2	47.6
5	ICB	6.95	7.52	19.6	32.4
6	ICB	1.82	2.02	18.5	22.2

```
library(tidyverse)
library(magick)

til8Plot <- ggplot(tilData,
                   aes(
                     x = group,
                     y = CD3CD8,
                     color = group
                   )) +
  geom_jitter(
    width = 0.1,
    size = 3
  ) +
  stat_summary(
    fun = mean,
    geom = "point",
    size = 5,
    color = "black"
  ) +
  stat_summary(
    fun.data = mean_sdl,
    fun.args = list(mult = 1),
    geom = "errorbar",
    width = 0.2,
    color = "black"
  ) +
  theme_classic() +
  labs(
    title = "CD3+CD8+",
    y = "CD3+CD8+(%)"
  ) +
  theme(plot.title = element_text(
    hjust = 0.5,
    face = "bold",
    size = 15
  ))
)
```

```
til8Plot
```



```
ggsave(
  filename = "CD8.pdf",
  plot = til8Plot,
  width = 8,
  height = 6
)
```

```
#Accessing the data folder
setwd("C:/Users/luisr/Documentos/Documents/Documentos - Copia/Mestrado/imunofenotipagem")

#Importing data
tilData <- read.csv("tils.csv")

head(tilData)
```

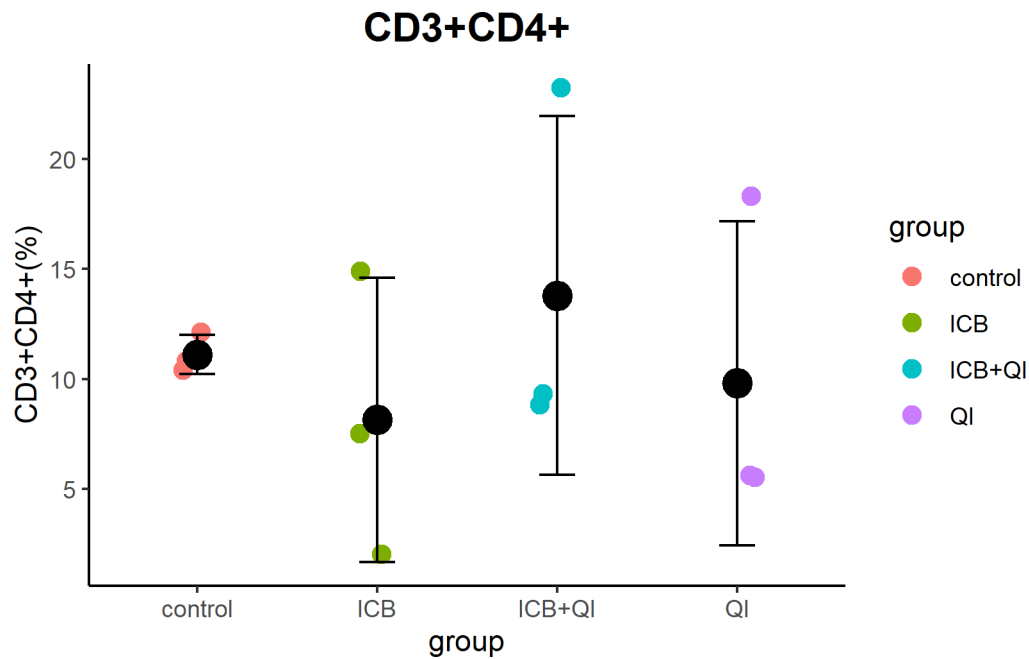
	group	CD3CD8	CD3CD4	CD3	exhaust
1	ICB+QI	22.90	23.20	34.6	28.6
2	ICB+QI	9.66	8.81	31.6	26.3
3	ICB+QI	9.13	9.35	27.4	22.0
4	ICB	11.60	14.90	26.2	47.6
5	ICB	6.95	7.52	19.6	32.4

6 ICB 1.82 2.02 18.5 22.2

```
library(tidyverse)
library(magick)

til4Plot <- ggplot(tilData,
                  aes(
                    x = group,
                    y = CD3CD4,
                    color = group
                  )) +
  geom_jitter(
    width = 0.1,
    size = 3
  ) +
  stat_summary(
    fun = mean,
    geom = "point",
    size = 5,
    color = "black"
  ) +
  stat_summary(
    fun.data = mean_sdl,
    fun.args = list(mult = 1),
    geom = "errorbar",
    width = 0.2,
    color = "black"
  ) +
  theme_classic() +
  labs(
    title = "CD3+CD4+",
    y = "CD3+CD4+(%)"
  ) +
  theme(plot.title = element_text(
    hjust = 0.5,
    face = "bold",
    size = 15
  ))
)
```

til4Plot



```
ggsave(
  filename = "CD4.pdf",
  plot = til4Plot,
  width = 8,
  height = 6
)
```

Now, ,let's plot exhaust TCD8+ lymphocyte

```
#Accessing the data folder
setwd("C:/Users/luisr/Documentos/Documents/Documentos - Copia/Mestrado/imunofenotipagem")

#Importing data
tilData <- read.csv("tils.csv")

head(tilData)
```

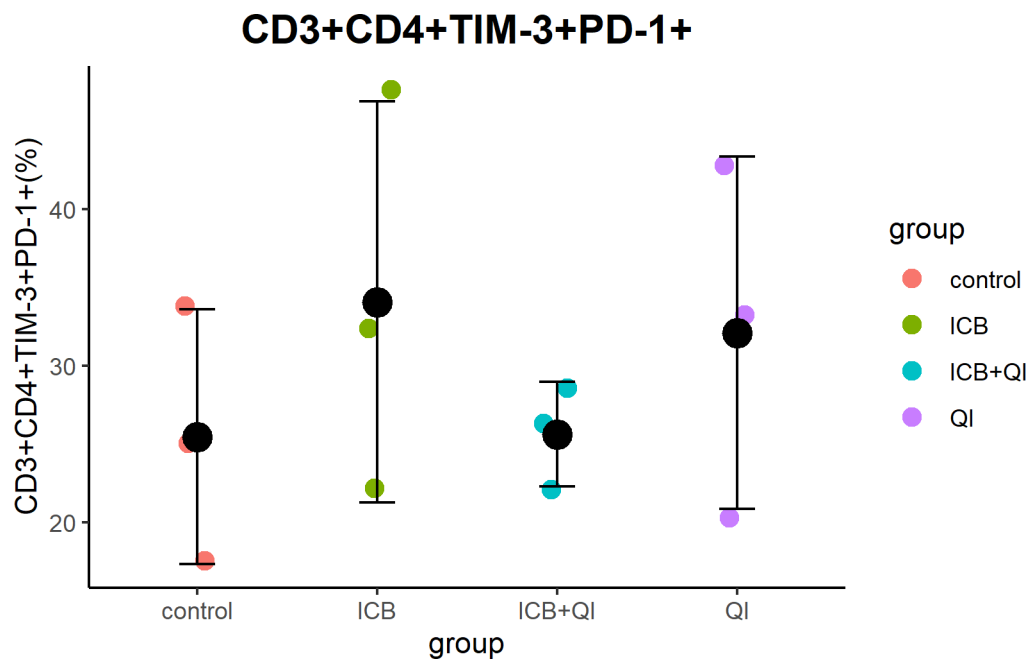
	group	CD3CD8	CD3CD4	CD3	exhaust
1	ICB+QI	22.90	23.20	34.6	28.6
2	ICB+QI	9.66	8.81	31.6	26.3
3	ICB+QI	9.13	9.35	27.4	22.0

4	ICB	11.60	14.90	26.2	47.6
5	ICB	6.95	7.52	19.6	32.4
6	ICB	1.82	2.02	18.5	22.2

```
library(tidyverse)
library(magick)

tilexPlot <- ggplot(tilData,
                    aes(
                      x = group,
                      y = exhaust,
                      color = group
                    )) +
  geom_jitter(
    width = 0.1,
    size = 3
  ) +
  stat_summary(
    fun = mean,
    geom = "point",
    size = 5,
    color = "black"
  ) +
  stat_summary(
    fun.data = mean_sdl,
    fun.args = list(mult = 1),
    geom = "errorbar",
    width = 0.2,
    color = "black"
  ) +
  theme_classic() +
  labs(
    title = "CD3+CD4+TIM-3+PD-1+",
    y = "CD3+CD4+TIM-3+PD-1+(%)"
  ) +
  theme(plot.title = element_text(
    hjust = 0.5,
    face = "bold",
    size = 15
  ))
)
```

tilexPlot



```
ggsave(  
  filename = "exhaust.pdf",  
  plot = tilexPlot,  
  width = 8,  
  height = 6  
)
```